Agenda

Wrap up generative models

Break

Nonlinearity

1. Generative Classification Models

1.1 Review

- Logistic Regression: $\log \frac{p(x)}{1-p(x)} = \beta_0 + x^T \beta$

- LDA $X \mid Y=k \sim N(\mu_k, \Sigma)$

- QDA $X \mid Y=k \sim N(\mu_k, \Sigma_k)$

- NB $X \mid Y=k$ are independent

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{k=1}^{K} \pi_k f_k(x)}$$

- $\hat{\pi}_k = \frac{n_k}{n}$

- $\hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} x_i$

- $\hat{\Sigma}$: pooled within-class covariance

- $\hat{\Sigma}_k$: within-class covariance

1.2 NB

$$p_k(x) = \frac{\pi_k f_{k1}(x_1) f_{k2}(x_2) \cdots f_{kp}(x_p)}{\sum_{k=1}^{K} \pi_k f_{k1}(x_1) \cdots f_{kp}(x_p)}$$

← works because we only care about this

1.3 Bernoulli NB: binary predictors

Y	x_1: free	x_2: link
S	0	1
S	1	1
S	1	1
S	1	1
NS	1	0
NS	0	1
NS	0	0

$$\hat{f}_{kj}(x_j) = \frac{1}{n_k} \sum_{i: y_i = k} x_{ji}$$

1.4 Multinomial NB - count predictors

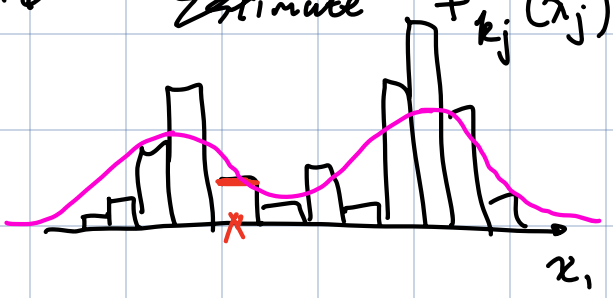
Text classification

1.5 Gaussian NB - predictors are Gaussian

$$f_{kj}(x_j) = N(\mu_{kj}, \sigma_{kj}^2)$$

Each predictor follows Normal distn, for each class, and are independent within classes

1.6 Estimate $f_{kj}(x_j)$ (histogram, Kernel density)



1.7 Boundary

Shape depends on $f_{kj}(x_j)$, but it's where

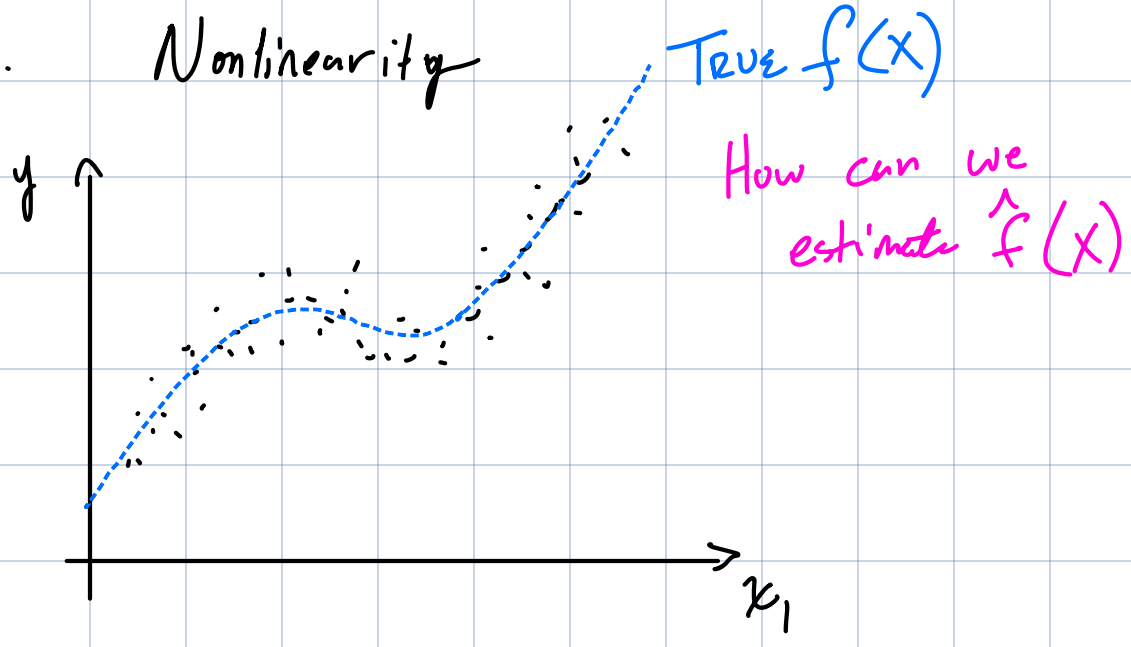
$p_k(x)$ are equal for each k

1.8 Summary

	LDA	QDA	NB	Logistic
Models:	$P(X\|Y)$	$P(X\|Y)$	$P(X\|Y)$	$P(Y\|X)$
Assumptions:	Gaussian shared Σ	Gaussian Σ_k	ind. predictors within classes	linear log-odds
Parameters:	moderate	most	few	$p+1$ fewest
Boundary:	linear	quadratic	additive	linear
Wins:	small n Gaussian	$n >> p$ covariances differ	n small p large	n large, assumption on others are doubtful

$K>2$:	fine	fine	fine	need multinomial logistic
Mixed predictor types:	NO	NO	YES	YES

2. Nonlinearity



2.1 Basis functions

Want to use least squares

$p=1$

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_J b_J(x_i) + \varepsilon_i$$

$$\underset{n \times J+1}{\mathbb{B}} = \begin{pmatrix} | & | & | & & | \\ 1 & b_1(x_i) & b_2(x_i) & \cdots & b_J(x_i) \\ | & | & | & & | \end{pmatrix}$$

$$\mathbb{y} = \mathbb{B}\beta + \varepsilon \qquad \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_J \end{pmatrix}$$

$$\hat{\beta} = (\mathbb{B}^T\mathbb{B})^{-1}\mathbb{B}^T\mathbb{y}$$

$$\hat{\mathbb{y}} = \mathbb{B}\beta \qquad \mathbb{y} = \mathbb{H}\mathbb{y} \qquad \mathbb{H} = \mathbb{B}(\mathbb{B}^T\mathbb{B})^{-1}\mathbb{B}^T$$

$$\hat{f}(x_0) = \hat{\beta}_0 + \sum_{j=1}^{J} \hat{\beta}_j \, b_j(x_0)$$

3 Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$